LoRA Meets Similarity Distribution Matching: Parameter-Efficient CLIP Tuning for Text-Based Person Search

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code on GitHub

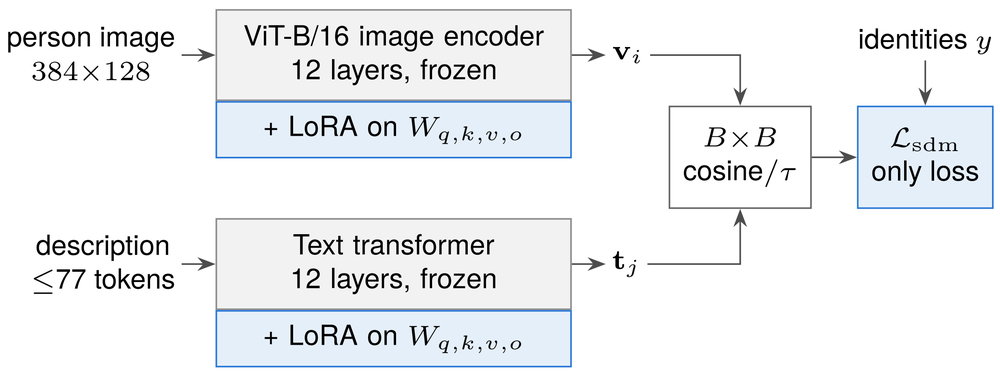
1 Motivation & Questions

Text-based person search ranks a gallery of pedestrian images by a free-form description. IRRA [1] fine-tunes all 149.6 M parameters of CLIP [2]. In federated learning every client uploads the trainable part in every round, so its size sets the communication cost. We ask two questions:

- How much accuracy does a *frozen* CLIP with LoRA [3] and the SDM loss alone give up, and how much does it save?
- Why does SDM beat InfoNCE? IRRA credits several positives per batch. We test this.

Existing (prior work)	Ours
CLIP ViT-B/16 [2]; LoRA [3]	Frozen CLIP + rank-4 LoRA on both encoders
SDM loss; IRRA results on RSTPReid [1,4]	SDM as the *only* loss; all LoRA runs
MAE robustness [5]	One-hot analysis of SDM; MPCE control; loss ablation

2 Method: LoRA-tuned CLIP, SDM only



Grey: frozen CLIP. Blue: the only trainable parameters.

Each attention projection becomes $W'\mathbf{x} = W\mathbf{x} + \frac{\alpha}{r}BA\mathbf{x}$ ($r = 4$, $\alpha = 8$): 96 modules, 491 520 trainable parameters (0.33%). No ID classifier, no IRR encoder. BA merges into W after training, so inference costs the same as CLIP.

3 Analysis: what SDM really optimises

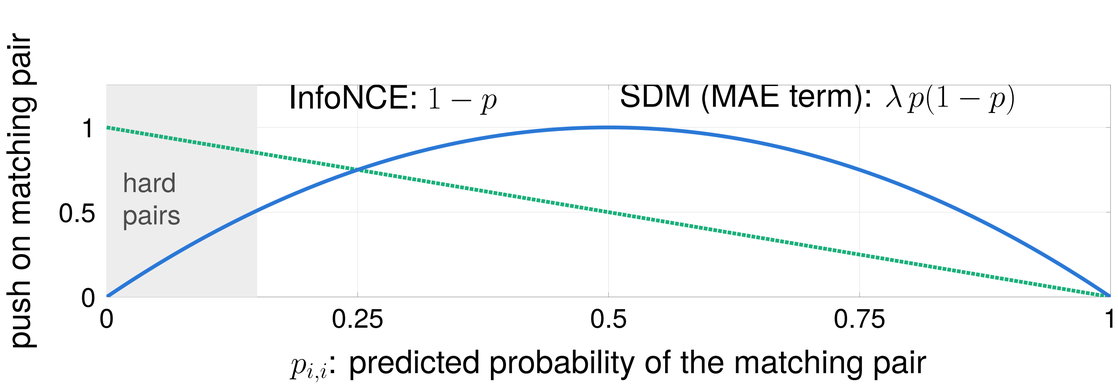
Step 1: a second positive is rare. IRRA motivates SDM with batches holding several examples of one person. RSTPReid has 10 per person among 37 010, so in a random batch of 64 only **1.52%** of rows have a second one: in 98.48% of rows SDM's target is one-hot, as in InfoNCE.

Step 2: the loss is still different. With a one-hot target, SDM becomes

$$\ell_i^{\text{sdm}} \approx \underbrace{\lambda\,(1 - p_{i,i})}_{\text{scaled MAE}} - \underbrace{H(p_{i,\cdot})}_{\text{entropy bonus}} \qquad \text{vs.} \qquad \ell_i^{\text{nce}} = -\log p_{i,i},$$

with $\lambda = \log(1/\epsilon) = 18.42$. MAE is bounded and robust to label noise [5].

Step 3: so the two losses train on different pairs.



Analytic, each curve scaled to peak 1. InfoNCE pushes hardest on the pairs it ranks worst, often ambiguous ones; SDM nearly ignores them.

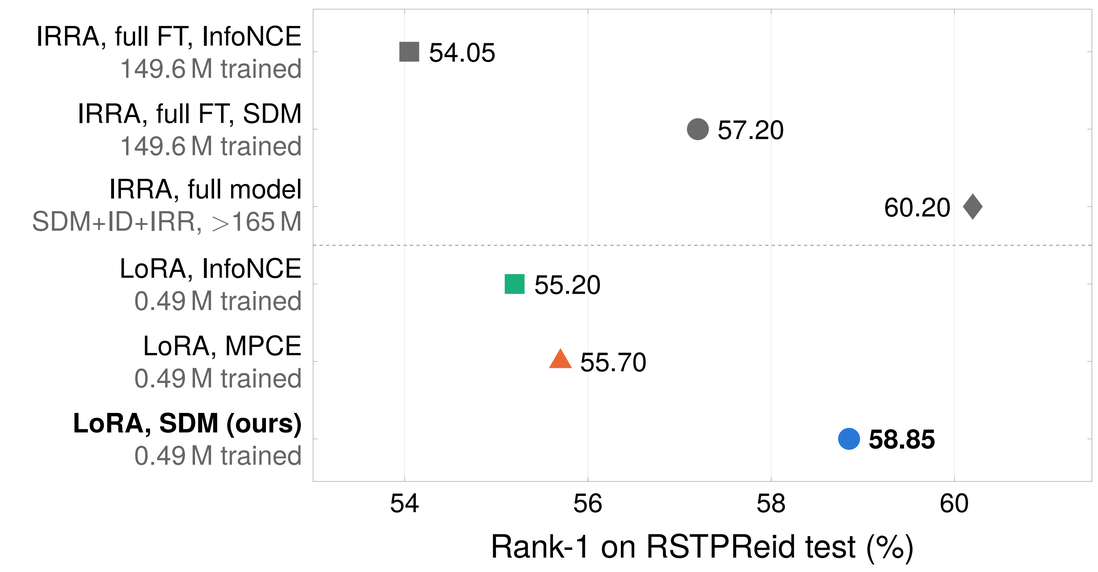
Test: MPCE keeps SDM's targets but uses InfoNCE's direction, $\mathrm{KL}(q\|p)$. If Step 3 is the reason, MPCE should behave like InfoNCE (block 5).

References

[1] D. Jiang, M. Ye. Cross-modal implicit relation reasoning and aligning for text-to-image person retrieval (IRRA). CVPR 2023.
[2] A. Radford *et al.* Learning transferable visual models from natural language supervision (CLIP). ICML 2021.
[3] E. J. Hu *et al.* LoRA: Low-rank adaptation of large language models. ICLR 2022.
[4] A. Zhu *et al.* DSSL: Deep surroundings-person separation learning for text-based person retrieval (RSTPReid). ACM MM 2021.
[5] A. Ghosh, H. Kumar, P. S. Sastry. Robust loss functions under label noise for deep neural networks. AAAI 2017.

4 LoRA vs. full fine-tuning

304× fewer trainable params · **0.49 M** trained vs. 149.6 M · **1.88 MiB** update vs. 571 MiB

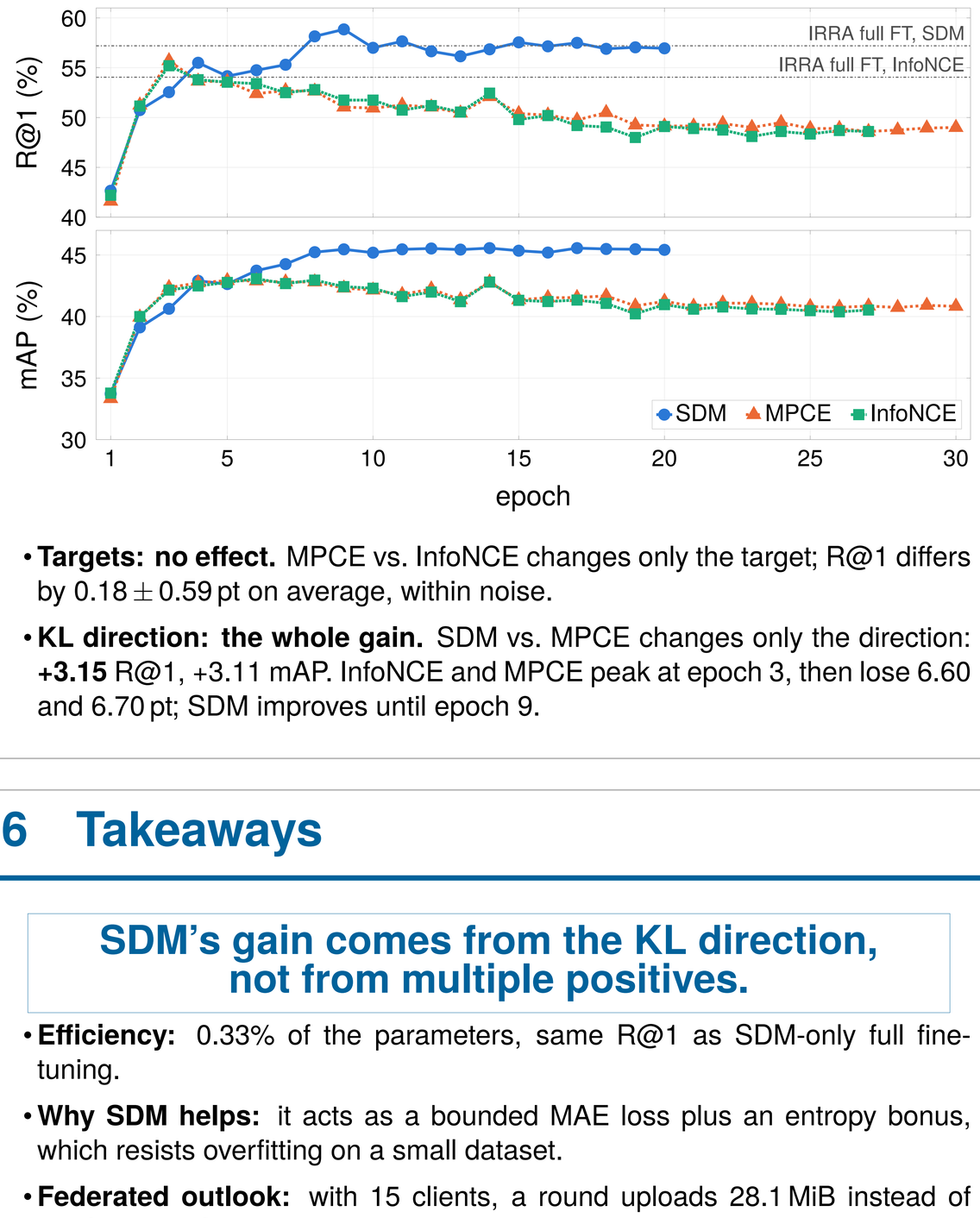


Peak R@1 (best test epoch, as IRRA reports). Grey: quoted from IRRA [1]. Ours: one seed each; standard error ≈1.1 pt.

LoRA+SDM is **on par** with SDM-only full fine-tuning: 58.85 peak and 57.18 ± 0.28 converged vs. 57.20. The complete IRRA model adds ID and IRR losses with their own parameters and is 1.35 pt higher.

5 Why does SDM help?

Loss	Target	Divergence	Peak R@1	Conv. R@1
InfoNCE	diagonal	cross-entropy	55.20	48.52
MPCE	identity	$\mathrm{KL}(q\|p)$	55.70	48.85
SDM	identity	$\mathrm{KL}(p\|q)$	**58.85**	**57.18**



- **Targets: no effect.** MPCE vs. InfoNCE changes only the target; R@1 differs by 0.18 ± 0.59 pt on average, within noise.
- **KL direction: the whole gain.** SDM vs. MPCE changes only the direction: **+3.15** R@1, +3.11 mAP. InfoNCE and MPCE peak at epoch 3, then lose 6.60 and 6.70 pt; SDM improves until epoch 9.

6 Takeaways

SDM's gain comes from the KL direction, not from multiple positives.

- **Efficiency:** 0.33% of the parameters, same R@1 as SDM-only full fine-tuning.
- **Why SDM helps:** it acts as a bounded MAE loss plus an entropy bonus, which resists overfitting on a small dataset.
- **Federated outlook:** with 15 clients, a round uploads 28.1 MiB instead of 8.4 GiB, and SDM has no identity-tied parameters to federate.

Limitations: one seed per run; IRRA numbers are quoted, not reproduced.